- Our uncertainty (measured by the predictive standard deviation) is close to zero around the three points in our training data, as demonstrated by the right panel. Going away from these data points, the standard deviation smoothly increases to one.

This means all of the nice properties of a GP such as smooth interpolation and uncertainty quantification are preserved when we go to higher dimensions.

This marks the end of Chapter 2. Overall, we have gained a conceptual understanding of what a Gaussian process is and learned how to implement a base GP model in Python using GPyTorch. As mentioned, we will dive deeper into the mean and covariance functions of a GP in Chapter 3, including their hyperparameters, and see how each of these components controls the behavior of our GP model.

2.5 Summary

- A multivariate Gaussian distribution models the joint distribution of many random variables. The mean vector denotes the expected values of the variables, while the covariance matrix models both the variances and the covariances among these variables.
- By applying Bayes' theorem, we can compute the posterior distribution of a multivariate Gaussian. Through this Bayesian update, variables that are similar to the observed variable are updated to reflect this similarity. Overall, similar variables produce similar predictions.
- A Gaussian process extends a multivariate Gaussian distribution to infinite dimensions, making it a distribution over functions. However, the behavior of a Gaussian process is still similar to that of a multivariate Gaussian distribution.
- Even without any training data, a Gaussian process may still produce predictions specified by the prior GP.
- Once trained on a data set, the mean prediction of a Gaussian process smoothly interpolates the training data points.
- One of the biggest advantages of using a Gaussian process is the calibrated quantification of uncertainty that the model offers: predictions around observed data points are more confident; predictions far away from training data, on the other hand, are more uncertain.
- Conditioning with a multivariate Gaussian distribution or a Gaussian process is visually similar to tying a knot at an observation. This forces the model to exactly go through the observation and reduces the uncertainty to zero.
- When using GPyTorch to implement Gaussian processes, we can write a model class that extends the base class in a modular way. Specifically, we implement two specific methods: `__init__()`, which declares the mean and covariance functions of the GP, and `forward()`, which constructs a multivariate Gaussian distribution for a given input.

2.6 Exercise

In this exercise, we will train a Gaussian process on a real-world data set that we saw in Chapter 1, which is shown again in Table 2.2.

Each data point (row) corresponds an alloy (a kind of metal) created by mixing lead (Pb), tin

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